Power-up

Initialize controller: `fcs_init()`
Zero controller state; set mode to DISARMED; clear armed/HOLD flags; initialize RC history to 1500 µs, except throttle, HOLD and flaps, which are initialized to 1000 µs.

Main loop
Wait for next 100 Hz tick; read RC frame and current time. In `fcs_run()`, use $\Delta t = 0.01$ s.

1. Validate the RC frame
Frame age: unsigned (`now_ms` − `stamp_ms`) $<$ 250 ms.
Every channel pulse must be between 880 and 2120 µs.

Frame valid?

Valid frame
Copy all channels to `rc_last`; set `rc_bad` $=0$; increment `rc_good`.

Invalid frame
Keep last good channel values; set `rc_good` $=0$; increment `rc_bad`.

Debounce link status
If `rc_ok` and `rc_bad` $\geq$ 10: set `rc_ok` = false.
Else if not `rc_ok` and `rc_good` $\geq$ 30: set `rc_ok` = true.
Otherwise retain the current link status.

2. Decode pilot inputs from last good frame
CH1 aileron, CH2 elevator, CH3 throttle, CH4 rudder, CH5 HOLD, CH6 flaps.
Aileron/elevator/rudder: normalize to $[-1, 1]$, apply deadband and expo; apply elevator sign. Throttle: normalize to $[0, 1]$.
HOLD switch is high if CH5 $>$ 1700 µs. Flaps: CH6 $<$ 1300: up; 1300–1699: half; $\geq$ 1700: full.

Not armed, link good, and throttle $<$ 0.05?

Set `armed` = true.

Keep the existing armed state.

3. Evaluate HOLD switch and latch state
If link bad, not armed, or HOLD switch low: clear HOLD.
If switch is low: also clear `hold_lockout`.
Otherwise, on a rising edge (previous switch low, current high), engage HOLD only if not locked out and throttle $\geq$ 0.15.
Latch `cmd_ail`, `cmd_ele`, current rudder stick, throttle stick, and flap selector.

HOLD active and |`st_ail`| $>$ 0.8 or |`st_ele`| $>$ 0.8?

Clear HOLD; set `hold_lockout` = true.
Toggle HOLD switch low to unlock.

Keep the current HOLD state.

Update `hold_sw_prev` to the current switch state.

RC link good?

FAILSAFE
Aileron = `fs_ail` = 0.10
Elevator = `fs_ele` = 0
Rudder = 0
Throttle = `fs_thr` = 0
Flaps = 0

Armed?

DISARMED
Aileron/elevator/rudder follow shaped sticks.
Throttle forced to zero.
Flaps follow the selector.

HOLD active?

HOLD
Use latched aileron, elevator, rudder and flaps.
Throttle = min(latched throttle, current throttle stick).

NORMAL
Aileron/elevator/rudder/throttle use shaped pilot inputs.
Flaps follow the selector.

Key configuration values
RC timeout: 250 ms
Link loss: 10 bad ticks; recovery: 30 good ticks
Throttle arm threshold: 0.05
HOLD engagement threshold: 0.15 throttle
HOLD breakout: 0.8 aileron/elevator deflection
Bank limit: ±45°
Pitch limits: $+20^\circ/-15^\circ$
Aileron–rudder mix: 0.25
Flap–elevator compensation: −0.05
Servo travel: 500 µs from centre

4. Surface authority limits
Aileron limit interpolates from 0.80 (flaps up) to 0.55 (full flaps), based on the previous output flap position.
Elevator clamped to $[-0.55, +0.70]$; rudder clamped to $[-0.85, +0.85]$.

Mode is DISARMED?

Reset `bank_est` and `pitch_est` to zero.

5. Dead-reckoned attitude limiters
Taper aileron near estimated bank limits of ±45°.
Taper elevator near estimated pitch limits of $+20^\circ$ and -15°.
Integrate assumed roll/pitch rates and self-levelling:

$$\dot{b} \approx k_b\,\mathrm{ail} - b/\tau_b, \qquad \dot{p} \approx k_p\,\mathrm{ele} - p/\tau_p$$

Save limited commands as `cmd_ail` and `cmd_ele` for future HOLD latching.

Important limitation
Bank and pitch are open-loop estimates, not sensor measurements. They can drift with wind, trim, CG and airspeed.

6. Control mixing
Target aileron = limited aileron command.
Target elevator = limited elevator $-\,0.05\times$ previous flap output.
Target rudder = clamp(rudder $+\,0.25\times$ aileron, −1, +1).
Target throttle = clamp(throttle $\times\,1.0$, 0, 1).
Target flap = selected flap command.

7. Clamp and slew-limit targets
Clamp target aileron/elevator to $[-1, +1]$.
Aileron/elevator/rudder slew: 8 units/s.
Throttle rise: 2 units/s; throttle fall: 3 units/s.
Flap slew: 0.4 units/s.

8. Apply servo reversals
Reverse aileron, elevator and rudder output if the corresponding configuration flag is enabled.

9. Convert normalized outputs to RC pulses
Aileron/elevator/rudder: 1500 + output × 500 µs.
Throttle/flaps: 1000 + output × 1000 µs.
CH5 status: 2000 µs if HOLD active; otherwise 1000 µs.

Write servo outputs

Next 100 Hz tick